CHAPTER

Parallel Imaging

6

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6.1 Introduction

Parallel imaging is an image reconstruction technique that relies on spatial information from an array of receiver coils to reduce aliasing artifacts arising from undersampled k-space data. Parallel imaging techniques typically exploit regular undersampling of k-space data, such that aliasing artifacts can be predicted and resolved using knowledge of the sensitivity profiles of the receiver coils. However, as we will see in later in this chapter, these techniques can also be extended to other more general acquisition schemes.

Parallel imaging is typically deployed to increase the speed of MRI scans without compromising spatial resolution. As discussed in previous chapters, there are many advantages of moving to accelerated data collection in MRI. Because many images with varied contrasts and orientations must be collected, clinical MRI protocols often take between 30 min to an hour, which can be excessive for sick or uncooperative patients. In addition to patient-comfort concerns, longer scan times for individual images can also negatively impact image quality. A longer scan increases the probability of patient motion during data collection, which can lead to motion artifacts. Furthermore, reducing scan times allows MRI to be used in applications that require high temporal resolution, such as cardiac, perfusion, and interventional imaging. In applications that are not time-sensitive, parallel imaging can be used to increase image resolution within a given scan duration. Parallel imaging can also offer improvements in image quality in sequences such as TSE and EPI by reducing the amount of data that must be collected in a single TR [1].

While there are many different parallel imaging-reconstruction algorithms, four main methods are frequently referenced in modern literature: SENSitivity Encoding (SENSE) [2], GeneRalized Autocalibrating Partially Parallel Acquisitions (GRAPPA) [3], iTerative Self-consistent Parallel Imaging Reconstruction (SPIRiT) [4], and Eigenvalue SPIRiT (ESPIRiT) [5]. Each of these methods will be discussed in this chapter, as well as variants for 3D [6] [7] [8], non-Cartesian [9] [10] [11], and dynamic imaging [12] [13] [14] [15] [16]. By the end of this chapter, the reader will ideally have attained a good understanding of the basic principles of parallel imaging reconstruction and will be able to describe the theory, advantages, and limitations of several state-of-the-art parallel imaging-reconstruction algorithms.

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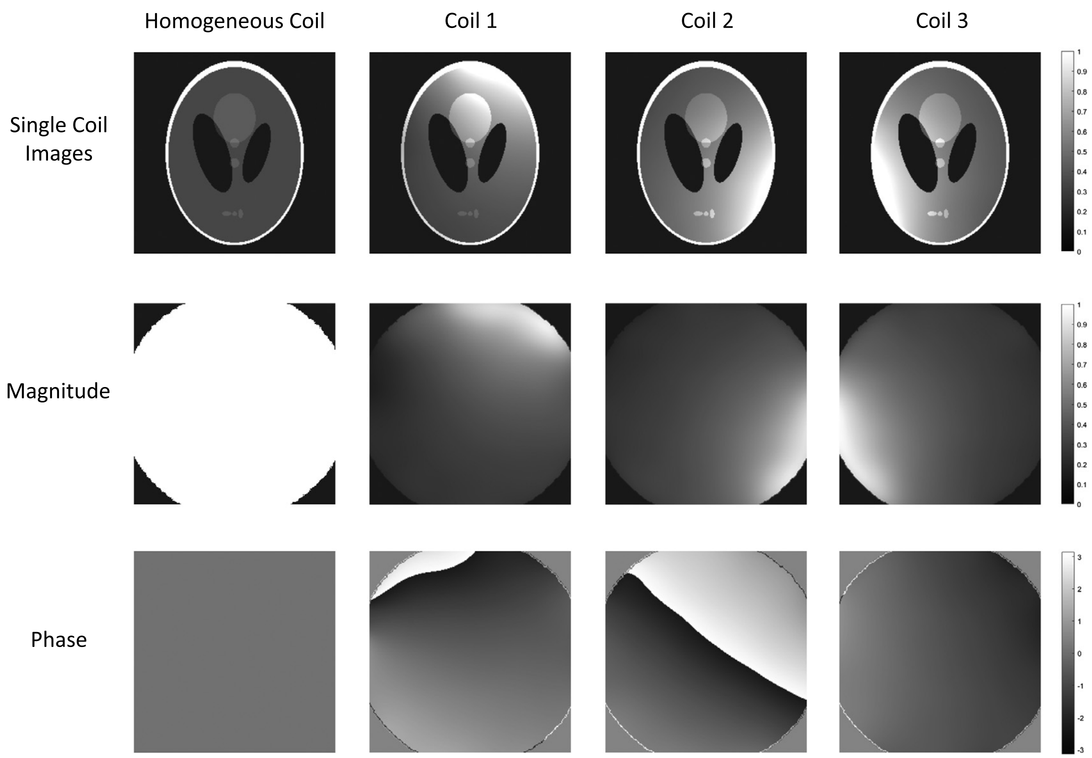

FIGURE 6.1

(*Top*) An image collected with various receiver coils, namely a coil with a homogeneous receive profile and no spatial variation in sensitivity (left), and three different elements of a receiver coil array placed at various locations around the object (center left, center right, and right). (Middle and bottom rows) The coil sensitivity profiles associated with these coils. When acquiring images with these coils, the localized sensitivity information provides an additional element of spatial localization that can complement gradient encoding. This effect can be seen in single coil images (top), where areas near to the coil appear brighter compared to those further away. Note that each coil sensitivity map is complex-valued and has a magnitude (center) and phase component (bottom). In the top row, lighter colors indicate that the coil is more sensitive to that portion of the object, and darker colors indicate both that there is lower sensitivity or that the object is not present (and thus there is no signal for the coil to be sensitive to).

Basic principles of parallel imaging

In MRI, data are collected in k-space using receiver coils. An array consisting of multiple receiver coils is often used to increase SNR, and an additional consequence of the use of such an array is that each individual coil is only sensitive to the portion of the body close to the coil. Fig. 6.1 shows an example of an image collected with a single homogeneous receiver coil, as well as images collected with several individual coils in a receiver array. Due to the local sensitivity patterns exhibited by the individual elements in the array, array coils can also provide some information for determining the spatial location of signals. In parallel imaging, this localized coil information is used to resolve aliasing artifacts present in undersampled images.

The sensitivity profile of a receiver coil can be described using a so-called "coil sensitivity map," as shown for the homogeneous receive coil and the individual elements of the array coil in Fig. 6.1. Coil sensitivity maps are complex-valued and are dependent on a variety of factors such as coil shape, placement, design, and even magnetic loading from the patient in the scanner. As a result, coil sensitivities are not constant and, in fact, can vary substantially from patient to patient. Parallel imaging techniques use a variety of methods for both estimating coil sensitivity information and applying it to reconstruct images.

The MR signal from a single receiver coil is given by the following equation:

$$S_j(\boldsymbol{k}) = \int_V C_j(\boldsymbol{r}) M_{xy}(\boldsymbol{r}) e^{(-2\pi i \boldsymbol{k}\cdot\boldsymbol{r})} d\boldsymbol{r} = \mathcal{F}\left\{C_j(\boldsymbol{r})\, M_{xy}(\boldsymbol{r})\right\} \tag{6.1}$$

where $\boldsymbol{k}$ is the k-space position vector, $\boldsymbol{r}$ is the spatial position vector, $\mathrm{S_j}(\boldsymbol{k})$ is the MRI signal acquired from coil j, $C_j(\boldsymbol{r})$ is the spatially varying coil sensitivity for coil j, and $M_{xy}(\boldsymbol{r})$ is the spatially varying magnetization. This equation can also be expressed in a discretized matrix form:

$$\boldsymbol{s} = \boldsymbol{BFCx} = \boldsymbol{Ex} \tag{6.2}$$

where $\boldsymbol{s}$ represents the MR signal, $\boldsymbol{B}$ represents the sampling operation, $\boldsymbol{F}$ represents the Fourier transform operation, $\boldsymbol{C}$ represents a matrix of discretized coil sensitivities, and $\boldsymbol{x}$ represents the magnetization at each point in the image. Frequently, the $\boldsymbol{BFC}$ operations are combined into the general MRI encoding matrix $\boldsymbol{E}$, which is used in many parallel imaging reconstruction methods.

When data are collected across multiple coils, the resulting single coil images each only contain part of the image, and they need to be combined in order to reconstruct the full image. There are a number of different methods for combining single coil images. Roemer, et al. [17] present an SNR-optimal recombination method in their original paper on multicoil acquisitions. Walsh, et al. [18] created a stochastic method for coil recombination for cases where coil sensitivity values are not exactly known. In cases where the application of these techniques is infeasible, the square root of the sum of squares of the single coil image values can be calculated pixel-wise to obtain an estimate for the true image.

There are a few terms and concepts that are common among all parallel imaging methods (and indeed most MRI reconstruction algorithms). Some of these concepts were already introduced in Chapter 2 and are briefly reviewed here for simplicity. The acceleration factor (Acc, also referred to here as R) describes the degree of undersampling for the acquired k-space data with respect to a fully-sampled k-space (as defined by the Nyquist–Shannon sampling theorem). In Cartesian acquisitions, undersampling is typically performed in the phase encoding direction since readout undersampling has minimal effect on image acquisition time. Therefore, when data are collected with an acceleration factor of $\mathrm{R} = 4$, this usually means that three out of every four lines of k-space are skipped in the acquisition. A diagram illustrating how k-space data are typically collected for a few different acceleration factors is shown in Fig. 6.2. The result of increasing the distance between collected k-space lines is that the image field-of-view is decreased; when parts of the object are outside of the smaller field-of-view, these parts appear in the smaller field-of-view on top of the parts which are still within the field-of-view, a phenomenon known as "aliasing." For example, when k-space data are accelerated by $\mathrm{R} = 2$, the field-of-view is decreased by a factor of two, and pixels which were originally half a field-of-view apart in

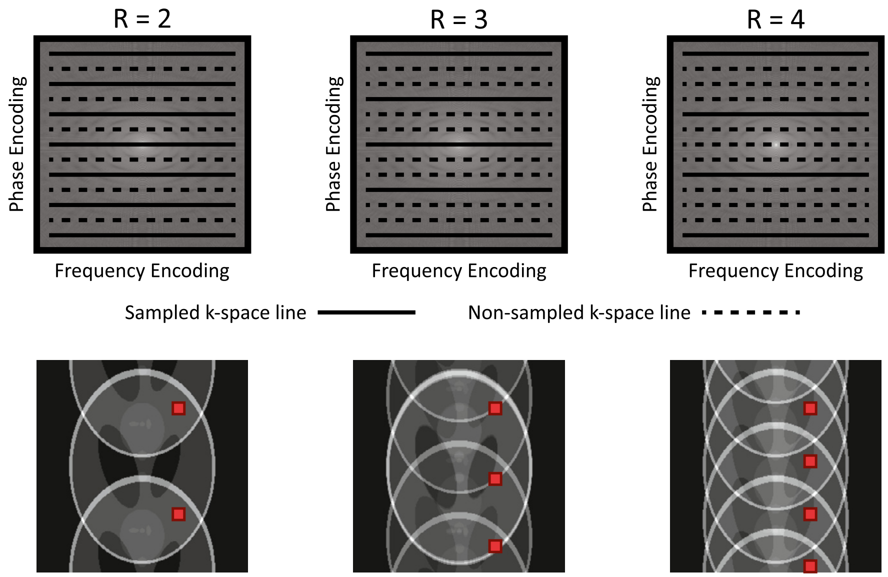

FIGURE 6.2

Sampling and aliasing patterns for various acceleration factors. **(top)** Uniform Cartesian sampling schemes for various acceleration factors. Note that the acceleration factor denotes the scan time reduction relative to a fully-sampled scan. **(bottom)** Aliasing patterns resulting from these undersampling schemes; the red (gray in print version) pixels in each image are examples of aliased points.

the unaccelerated image fall on top of one another (see Fig. 6.2, bottom left). The higher the acceleration factor, the larger number of pixels that fold together in the image, and the more challenging it can be to use the image for clinical or research purposes.

As a theoretical limit, the acceleration factor which can be resolved using parallel imaging is constrained by the number of receiver coils used to collect the data and their geometry with respect to the direction of undersampling. A coil array with completely independent coil sensitivity profiles with respect to the undersampling direction should in theory be able to reconstruct data with an acceleration factor equal to the coil count. However, it is rarely possible to resolve aliasing artifacts using parallel imaging when the acceleration factor is as high as the number of receiver coils in the array. The practical limits on the acceleration factor depend on the type of scan being performed and what tradeoffs can be made between image quality and temporal resolution. If there is sufficient SNR, parallel imaging can be used to accelerate any kind of clinical scan to improve patient compliance and comfort, even if there is no anticipated motion. In such a setting, modest acceleration factors of $R = 2$ are used to ensure high quality images. For specific applications, such as cardiac or perfusion scans, where both information concerning the motion of anatomical structures as well as temporal changes are needed clinically, acceleration factors of three or four are common. Some research scans can have acceleration factors as high as $R = 16$, depending on the size of coil arrays used in the scan. However, such high acceleration rates are only possible with special receiver coil arrays and with advanced reconstruction techniques that are not yet appropriate for widespread clinical adoption.

Another important concept in parallel imaging is how robustly the parallel imaging system can be solved, and thus how well the reconstructed image recapitulates the desired image. As explored later in this chapter, parallel imaging methods estimate missing data by setting up systems of equations and then solve these equations for the values of the unknown nonsampled points. While parallel imaging reconstruction algorithms vary in how these equations are set up and what they represent, all algorithms rely on additional calibration information, which is used to solve these systems of equations. The quality of the reconstructed images is highly dependent on the factor by which the system is overdetermined, which in turn depends on the quality and amount of calibration information available. Reconstructions with equations that are more overdetermined tend to be more robust and less sensitive to noise, but require more computational power to solve and a longer acquisition time to collect this information.

Finally, parallel imaging impacts the SNR of the resultant image [19]. The decrease in SNR can be calculated based on the following equation:

$$SNR_{PI} = \frac{SNR_0}{g\sqrt{R}}, \tag{6.3}$$

where SNR_{PI} is the SNR of the parallel imaging reconstruction, SNR_0 is the SNR of an equivalent fully-sampled image, R is the acceleration factor, and g is the coil geometry factor, or g-factor. The g-factor is dependent on both the aliasing patterns in the accelerated images and the coil sensitivity profiles. The g-factor is (almost) always greater than or equal to one as parallel imaging does not usually improve the SNR of an image. The g-factor is spatially dependent and can be represented as a g-factor map. Explicit knowledge of the g-factor is not required for parallel imaging reconstructions, but it is frequently used to analyze the performance of a parallel imaging technique.

6.2 Fundamental techniques

SENSE and GRAPPA are the most commonly used parallel imaging techniques due to their robustness in a wide range of imaging scenarios. SENSE is an image-domain technique that uses explicit knowledge of the coil sensitivity maps to solve for unaliased pixel values. GRAPPA is a k-space based technique that extracts coil information from a fully-sampled calibration space and reconstructs missing points in k-space. They also form the theoretical framework for other approaches such as CG-SENSE, SPIRiT, and ESPIRiT, and are the basis for a variety of techniques for dynamic data sets, such as k-t SENSE, TGRAPPA, k-t GRAPPA, and PEAK-GRAPPA, all of which are discussed later in this chapter.

Sensitivity encoding (SENSE)

Siemens: mSENSE; GE: ASSET; Phillips: SENSE

SENSE [2] is one of the earliest proposed and most intuitive parallel imaging techniques. SENSE operates in the image domain and uses the principle that each point in an aliased image is the sum of several values in the unaliased image weighted by the local coil sensitivity at each point. This principle is shown in Fig. 6.3. If the coil sensitivity maps are known and enough independent coils are used for data collection, a system of equations can be solved to find the unknown image values.

A simple implementation of the SENSE algorithm for uniformly undersampled Cartesian data considers a set of equations for each point in the aliased image and for each receiver coil used for acquisition. For example, for an acceleration factor of $R = 3$, and data collected with three coils ($N_c = 3$),

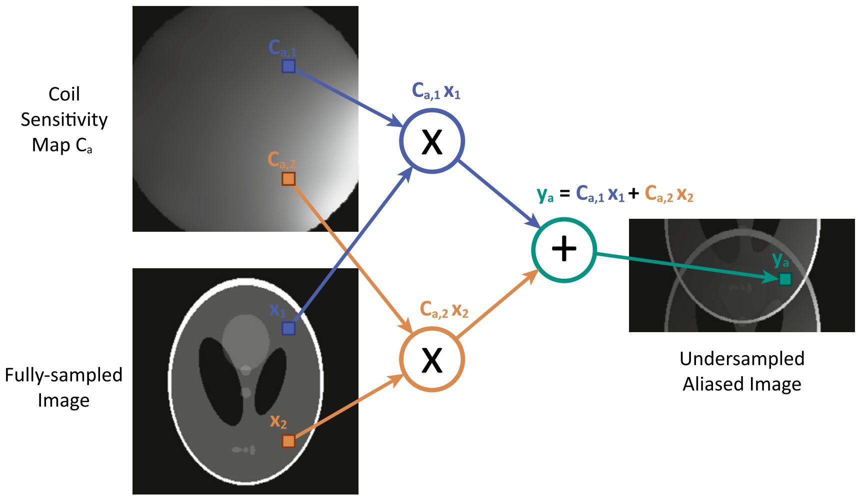

FIGURE 6.3

Visual representation of the SENSE equation for R = 2. The aliased image is the sum of two image pixels, weighted by the appropriate coil sensitivity at each point. This equation is written for each aliased single coil image to define a system of equations that is solved to reconstruct both aliased pixels.

the matrix equations for each pixel would be:

$$\begin{bmatrix} y_a \\ y_b \\ y_c \end{bmatrix} = \begin{bmatrix} c_{a1} & c_{a2} & c_{a3} \\ c_{b1} & c_{b2} & c_{b3} \\ c_{c1} & c_{c2} & c_{c3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \tag{6.4}$$

Here, y_a denotes the aliased pixel in coil a; y_b and y_c the aliased pixels in coils b and c; x_1, x_2, x_3 are the pixels in the unaliased image to be recovered; and the c elements represent the coil sensitivity value for the coil indicated by the letter subscript at the location indicated with the numerical subscript. This implementation is shown visually in Fig. 6.4. This equation can be written more compactly in matrix form:

$$\boldsymbol{y} = \boldsymbol{C}_p \boldsymbol{x} \tag{6.5}$$

where $\boldsymbol{y}$ is a vector of length N_c containing the aliased pixels for each coil, $\boldsymbol{C}_p$ is an $N_c \times R$ matrix containing the coil sensitivity values for this set of pixels, and $\boldsymbol{x}$ is the unknown but desired vector of R unaliased image pixels. As the values of the aliased pixels are known and the values of the coil sensitivity maps can be determined as subsequently described, the goal is to solve for the R unknown image pixels in $\boldsymbol{x}$. This matrix equation can be solved by applying the inverse of the coil sensitivity encoding matrix $\boldsymbol{C}$ to both sides of the equation:

$$\hat{\boldsymbol{x}} = \boldsymbol{C}_p^{\dagger} \boldsymbol{y}, \tag{6.6}$$

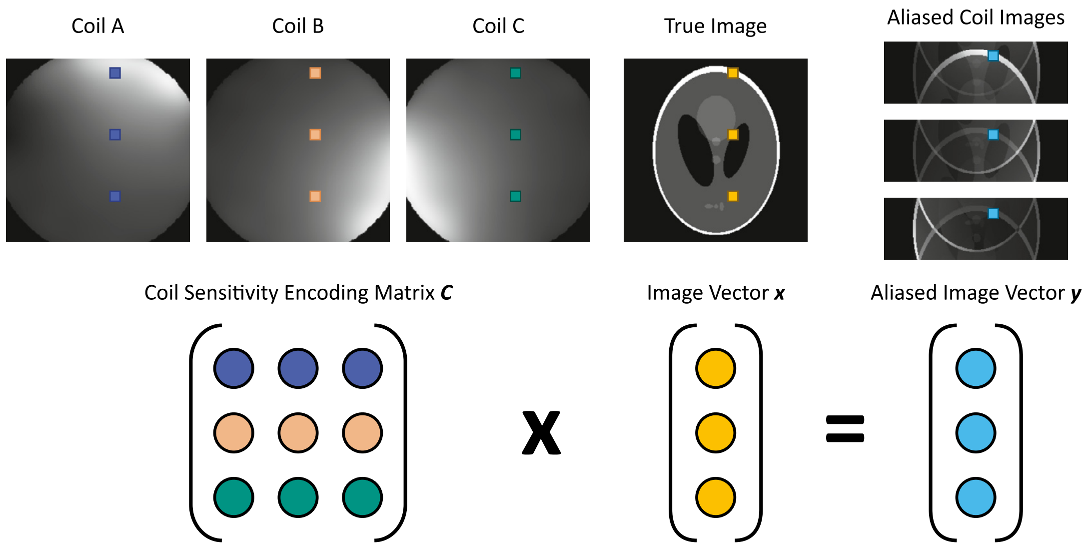

FIGURE 6.4

Matrix representation of the SENSE reconstruction. Coil sensitivity values are placed as row vectors in the coil sensitivity encoding matrix $\boldsymbol{C}$. The true image vector $\boldsymbol{x}$ is related to the pixels in the aliased images $\mathbf{y}$ through the application of this sensitivity encoding matrix, as can be seen in Eq. (6.2).

where the † symbol indicates the matrix pseudoinverse. By solving this equation, a single set of aliased points can be unfolded and placed in the appropriate locations (1, 2, and 3, as previously noted). A similar system of equations must be solved for each set of aliased pixels in order to reconstruct the full image.

At higher acceleration factors, more pixels are aliased together, and thus the length of the vector of the unknown original pixel values, $\boldsymbol{x}$, increases, and additional columns are added to the coil sensitivity matrix. Theoretically, for the SENSE equations to be solvable, the acceleration factor can be at most equal to the number of coils. In practice, it is usually much smaller given the fact that coils are not perfectly independent (increasing the condition number of $\boldsymbol{C}_p$ and thus leading to noise amplification when solving the inversion), and image SNR is reduced for higher acceleration factors. The need to solve for more unknown quantities in each system of equations makes each system less overdetermined, which can negatively impact the quality of the reconstruction. If more (independent) coil elements are used to collect data, more rows are added to the system, making it more overdetermined and robust to noise in coil sensitivity measurements.

As just described, the crucial information required for SENSE reconstruction are maps of the coil sensitivity profiles. A simple method for estimating coil maps is described in the original SENSE paper [2]. Images of the same object can be collected using both the body coil (with a homogeneous sensitivity profile) and the receiver array to be used for parallel imaging. The images collected with the array coils are then divided by the body coil image, yielding a rough estimate for the coil sensitivities at each pixel. These coil sensitivity maps can be thresholded to focus on the area of interest and denoised to reduce

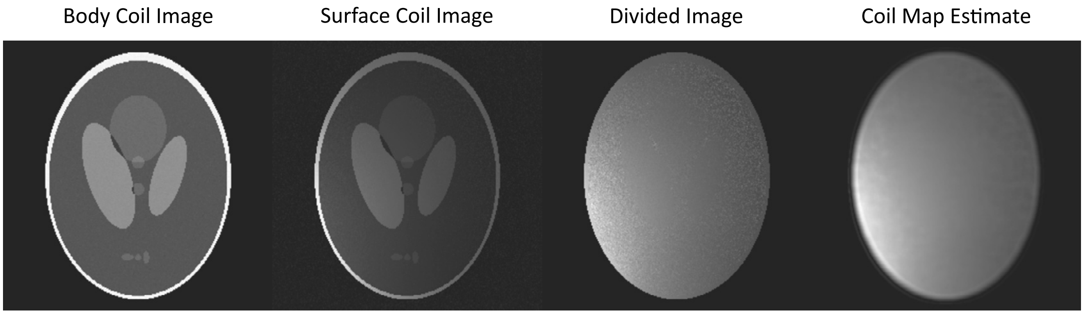

FIGURE 6.5

Example process for calculating coil sensitivity maps. An image collected using an element of the array coil is divided pixel-wise by an image collected using the homogeneous body coil. This produces a noisy estimate of the coil sensitivity map for that array coil element. This initial estimate can be filtered with a 2D polynomial smoothing function to yield a smoother estimate for the coil sensitivity.

noise or other spurious signal that could negatively impact the SENSE reconstruction. This method for estimating coil maps is shown in Fig. 6.5. Other methods for estimating coil maps include adaptive reconstruction [18] and ESPIRiT [5].

As described in Eq. (6.1), the amount of noise present in the reconstruction is related to the g-factor. For SENSE reconstructions, the g-factor can be calculated pixel-wise over the aliased image using the following equation:

$$g_i = \sqrt{[\left(\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{C}_p\right)^{-1}]_{i,}[\left(\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{C}_p\right)]_{i,i}}, \tag{6.7}$$

where $^{\mathrm{H}}$ indicates the matrix conjugate transpose, $^{-1}$ indicates the matrix inverse, and i is the index of the pixel of the original image that aliases onto the point in the aliased image. Note that the i, i subscripts indicate the i^{th} diagonal element of the $\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{C}_p$ or $(\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{C}_p)^{-1}$ matrices. A sample g-factor map for a SENSE reconstruction is shown in Fig. 6.6.

The original SENSE implementation also includes a receiver noise covariance matrix term that describes noise levels for a given pixel and how the noise is correlated across coils. The reconstruction equation (Eq. (6.6)) can be restated to include these terms as well:

$$\hat{\boldsymbol{x}} = (\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{\Psi}^{-1}\boldsymbol{C}_p)^{-1}\boldsymbol{C}_p^{\mathrm{H}}\boldsymbol{\Psi}^{-1}\boldsymbol{y}, \tag{6.8}$$

where $\boldsymbol{\Psi}$ is the receiver noise covariance matrix. In the case where the noise is independent and constant across coils, this matrix is the identity matrix, and this equation simplifies to Eq. (6.4). Including the $\boldsymbol{\Psi}$ term effectively weights the reconstruction matrix so that coils with a higher SNR have more influence on the reconstructed image vector [20], which can reduce errors due to low signal coils.

The SENSE reconstruction method has a few drawbacks. Most notably, it requires accurate coil sensitivity maps to work properly. Inaccuracies, misregistration, or noise in the coil maps can propagate into the reconstructed image, causing errors in reconstruction. In addition, subject motion between

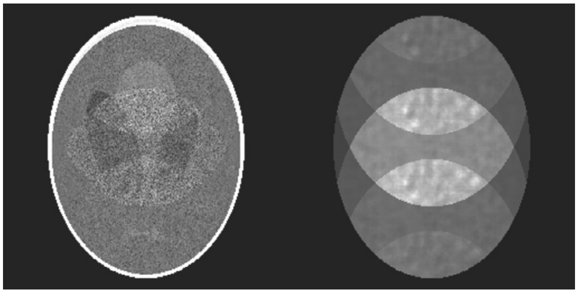

FIGURE 6.6

A sample reconstructed image (left) and g-factor map (right) from a SENSE reconstruction with R = 4. Note the noisy regions in the reconstructed image that correspond to regions with high g-factor values.

coil sensitivity estimation and data acquisition can further corrupt reconstruction quality. The straightforward SENSE implementation described here is also limited to uniformly sampled Cartesian data. The more generalized CG-SENSE algorithm (described later in this chapter) is required to reconstruct images with more complex sampling schemes.

SENSE and related techniques have been used for a wide variety of clinical applications. For example, in the heart, it has been used for cine imaging [21] [22] and myocardial perfusion [23]. In the brain, it has been used to accelerate fMRI scans [24] [25]. SENSE has also been applied in abdominal imaging for renal perfusion scans [26] and abdominal angiography [27], among many other applications.

Generalized autocalibrating partially parallel acquisitions (GRAPPA)

Siemens: GRAPPA; GE: ARC

Unlike SENSE, which unfolds aliased pixels in the image domain, the GRAPPA algorithm is a k-space-based technique. In GRAPPA, the unsampled k-space points are estimated by exploiting relationships between neighboring points [3]. GRAPPA requires a small fully sampled portion of the dataset for calibration (often referred to as the Autocalibration Signal, or ACS). From the ACS data, a GRAPPA weight set can be calculated to capture the relationship between neighboring k-space points. This weight set is then applied to the remaining undersampled data in order to estimate the missing k-space points. Finally, a Fourier transform is used to generate the unaliased reconstructed image from the synthesized k-space data.

The theory behind GRAPPA can be best understood by reexamining the idea of SENSE in k-space. As just described, the individual single coil images acquired with MRI can be imagined as the true image multiplied element-wise by a coil sensitivity map, as shown in Eq. (6.2). In k-space, this process corresponds mathematically to convolving the Fourier transform of the coil sensitivities with the Fourier transform of the image:

$$\mathcal{F}\left\{C(\boldsymbol{r})\, M_{xy}(\boldsymbol{r})\right\} = \mathcal{F}\{C(\boldsymbol{r})\} * \mathcal{F}\left\{M_{xy}(\boldsymbol{r})\right\}. \tag{6.9}$$

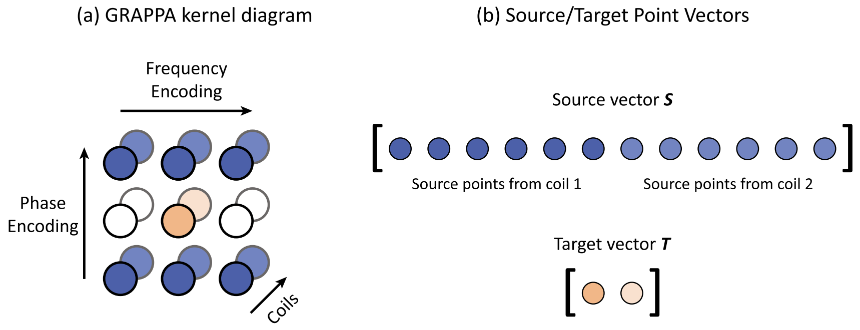

FIGURE 6.7

(a) A sample GRAPPA kernel, with source points (blue (dark gray in print version)) and target points (orange (light gray in print version)), on a data set with an acceleration factor of R = 2. This is a 3 × 2 kernel, referring to the number of source points in the kernel that extend over three readout points and two phase-encoding lines. **(b)** Each set of source and target points from a kernel forms one row vector of data. Row vectors from multiple kernels can be concatenated vertically to form the source and target matrices for reconstruction.

The result of this convolution is that each point of the measured k-space signal contains some information about the missing k-space points; in other words, there is a mathematical relationship between neighboring points in k-space due to the inhomogeneous coil sensitivities. In any small neighborhood of points, these relationships contain mostly structural information about the image. However, when looking at many groupings of local points, the local structural information averages out, and the convolution pattern due to the coil sensitivity can be elucidated.

At the core of the GRAPPA implementation is the GRAPPA kernel, a recurring pattern of neighboring points in k-space. The kernel is made up of source and target points; in an undersampled dataset, the source points are collected, but the target points are unknown and must be reconstructed. A diagram of a typical kernel made up of several source points for each target point is shown in Fig. 6.7; datapoints along the fully sampled phase-encoding lines above and below the target point are often used as the source points. GRAPPA kernels also extend in the coil dimension because data from every coil is used to derive GRAPPA weights and reconstruct target points across all coils.

The mathematical relationship between points in a kernel is captured in the GRAPPA weight set. For kernels in the fully sampled ACS regions, the source points and target points are known, allowing the relationship between these points to be derived. When data are collected along a standard uniform Cartesian trajectory, the ACS data typically take the form of a few readout lines near the center of k-space, and the rest of k-space is undersampled. A diagram of this acquisition scheme with an acceleration factor of R = 2 is shown in Fig. 6.8. The center of k-space is often used as the ACS since this region contains high signal data, reducing the influence of noise on the GRAPPA weights. To determine the GRAPPA weights, source points and target points are collected for each occurrence of the kernel in the ACS data. The number of kernel repetitions in an ACS is given by the following equation:

$$(N_R - s_R + 1)(N_{PE} - R\,(s_{PE} - 1)), \tag{6.10}$$

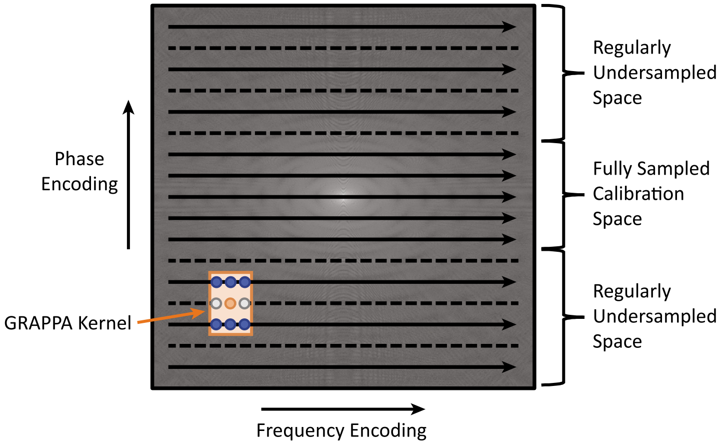

FIGURE 6.8

Sample k-space acquisition scheme and GRAPPA kernel for Cartesian GRAPPA.

where N_R and N_{PE} are the size of the ACS in frequency- and phase-encoding directions, and s_R and s_{PE} are the size of the GRAPPA kernel. For an ACS that is made up of 256 readout points and 9 phase-encoding lines, an acceleration factor of R = 2, and a kernel of size 3 × 2, there are 1778 repetitions of the kernel. These kernel repetitions are rearranged into source and target matrices of size (# of kernel repetitions in ACS) × (# of source or target points in kernel × number of coils). The relationship between the source and target matrices is expressed by the following equation:

$$\boldsymbol{T}_{ACS} = \boldsymbol{S}_{ACS}\boldsymbol{W}, \tag{6.11}$$

where $\boldsymbol{T}$ is the target point matrix, $\boldsymbol{S}$ is the source point matrix, and $\boldsymbol{W}$ is the GRAPPA weight set. In order to solve for the weight set, the pseudoinverse of the source point matrix is applied to both sides of the equation, yielding:

$$\boldsymbol{W} = \boldsymbol{S}^{\dagger}_{ACS}\boldsymbol{T}_{ACS}. \tag{6.12}$$

Once the GRAPPA weights have been determined, the target points in the undersampled k-space data can be recovered by applying these weights to source point matrices constructed using the same kernel structure:

$$\boldsymbol{T} = \boldsymbol{S}\boldsymbol{W}. \tag{6.13}$$

The amount of ACS data required for the GRAPPA reconstruction is related to the size of the GRAPPA kernel and number of coils. The system in Eq. (6.8) must be overdetermined for an accurate reconstruction, and so the number of kernel repetitions in the ACS must be greater than the number of source points in a kernel across all coils.

In addition, there are a few different options for collecting ACS data. First, the contrast of the ACS data does not have to match the rest of the scan, which can be beneficial for sequences with long repetition times. In addition, ACS data can be collected either during the regular scan, or as a separate scan either before or after undersampled data acquisition.

The main advantage of GRAPPA is that it is relatively robust in a variety of clinical imaging scenarios. One of the main reasons for this is that the GRAPPA weights are determined using many ACS kernels, making the system more robust against poor data in any specific kernel and ultimately allowing for high-quality reconstructions in nonideal conditions [1] [28]. GRAPPA also does not require coil sensitivity maps to be explicitly formulated, which can also be a source of reconstruction error [29].

GRAPPA has been used clinically and in a wide variety of research applications, including cardiac cine imaging [22], myocardial perfusion [30], interventional catheter placement [31], spinal TSE imaging [32], and MR angiography [33].

6.3 Advanced techniques

In addition to SENSE and GRAPPA, there are many more advanced parallel imaging methods which work along the same basic principles but which can improve the robustness of the reconstructions or be used to accelerate Cartesian and non-Cartesian data acquisitions with more general sampling patterns. These techniques pose the reconstruction problem within an optimization framework, which can be solved using iterative methods, such as the conjugate gradient algorithm described in Chapter 3, and can also incorporate other regularization options as desired to improve image quality.

Conjugate gradient SENSE (CG-SENSE)

Conjugate Gradient SENSE [9] is a generalization of the SENSE algorithm designed to work with data collected along nonuniform Cartesian and non-Cartesian trajectories. When Cartesian data are undersampled with uniform or regular sampling (e.g., every two lines in k-space), the aliasing artifacts are predictable and can be easily modeled, a feature exploited by SENSE to unfold small groups of aliased pixels. With nonuniform Cartesian (e.g., random Cartesian undersampling) and non-Cartesian (e.g., radial/spiral) sampling, the aliasing patterns are more complex; each pixel can affect every other pixel in the resulting aliased image. Solving this problem using the SENSE approach described before would require a very large coil sensitivity encoding matrix of size (total number of k-space signals) × (image matrix size). Due to its size (and typically ill-conditioned nature), the pseudoinverse of this matrix cannot be easily calculated.

To solve this problem and enable SENSE to be used to reconstruct images collected with undersampled nonuniform Cartesian and non-Cartesian data, we begin with the discretized signal model from Eq. (6.2):

$$\boldsymbol{E}\boldsymbol{x} = \boldsymbol{s}. \tag{6.14}$$

The generalized encoding matrix $\boldsymbol{E}$ corresponds to a multichannel version of the encoding matrix described for a single coil in Chapter 2. For the specific case of non-Cartesian data, $\boldsymbol{E}$ corresponds to a multichannel version of the nonuniform fast Fourier transform (NUFFT) operations: coil multiplication, Fourier transform, and nonuniform sampling. To reconstruct the image, the pseudoinverse of the encoding matrix is applied to the k-space data:

$$\hat{\boldsymbol{x}} = (\boldsymbol{E}^{\mathrm{H}}\boldsymbol{E})^{-1}\boldsymbol{E}^{\mathrm{H}}\boldsymbol{s}, \tag{6.15}$$

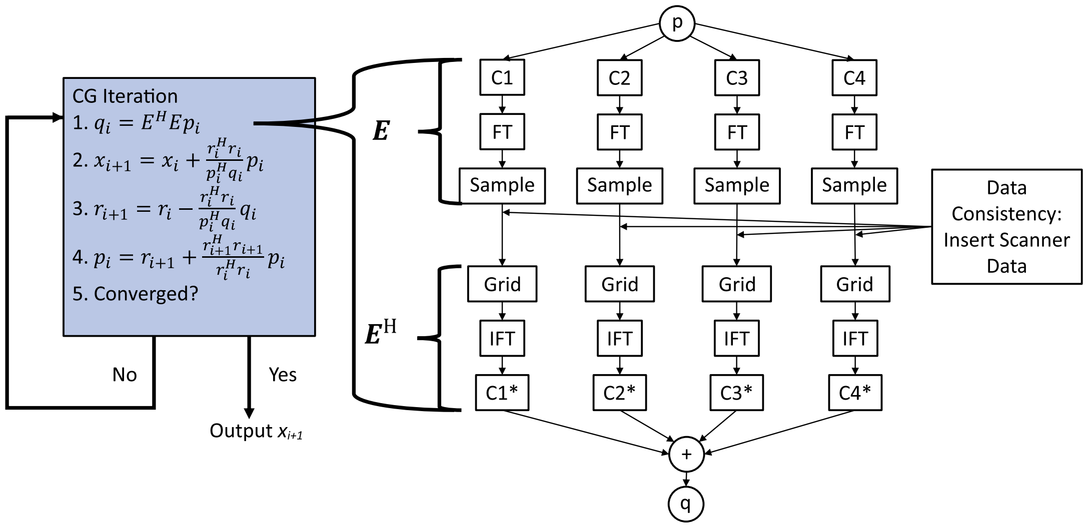

FIGURE 6.9

Diagram of the CG-SENSE algorithm. At the core of this algorithm is the conjugate gradient (CG) algorithm (blue (mid gray in print version) box). The $\boldsymbol{E}^H\boldsymbol{E}$ operation is shown in detail and is broken down into coil multiplication, Fourier transform, re-sampling, data insertion, gridding (in the case of non-Cartesian sampling), inverse Fourier transformation, and coil combination steps.

where $\hat{\boldsymbol{x}}$ is the reconstructed image, $\boldsymbol{s}$ is the k-space signal data, $^{\mathrm{H}}$ indicates the matrix conjugate transpose, and $^{-1}$ indicates the matrix inverse. Since $\mathbf{E}$ is a very large matrix, inverting the matrix $\boldsymbol{E}^H\boldsymbol{E}$ is computationally intensive. However, this matrix inversion can be eliminated by multiplying both sides of the equation by $\boldsymbol{E}^H\boldsymbol{E}$:

$$\boldsymbol{E}^{\mathrm{H}}\boldsymbol{E}\hat{\boldsymbol{x}} = \boldsymbol{E}^{\mathrm{H}}\boldsymbol{s}. \tag{6.16}$$

CG-SENSE uses the conjugate gradient algorithm to iteratively solve this equation. The implementation of the conjugate gradient algorithm for CG-SENSE in shown in Fig. 6.9. CG-SENSE slightly deviates from the standard CG implementation discussed in Chapter 3 by including a data consistency step, where data collected at the MRI scanner are inserted into the points that were originally sampled to ensure that the final solution image is consistent with the non-Cartesian data acquired.

While CG-SENSE is more computationally efficient than inverting the generalized sensitivity-encoding matrix, it still requires multiple NUFFT operations in the case of non-Cartesian trajectories that are relatively computationally intense. In addition, like SENSE, CG-SENSE requires accurate coil sensitivity maps. Errors in these coil sensitivity maps, density compensation required for the NUFFT, or intensity weighting can become exaggerated after iterating through these operations multiple times.

Another benefit of the CG-SENSE algorithm is that regularization terms can easily be incorporated within the CG iterations, thereby enforcing desirable qualities over the entire reconstructed image or k-space signal. Some examples include using Tikhonov regularization to reduce noise [34] or using L1-regularization of the wavelet transform of the image for simultaneous compressed sensing recon-

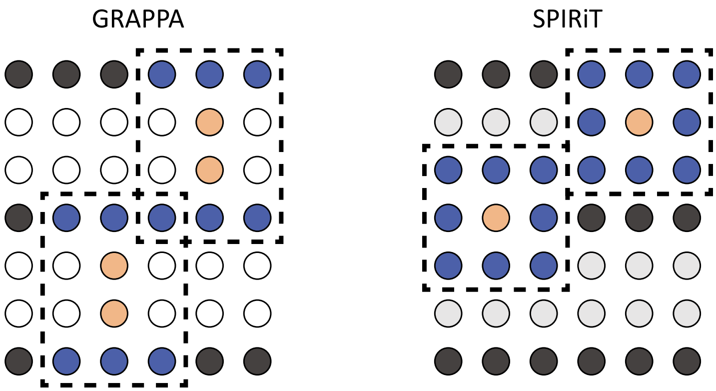

FIGURE 6.10

Differences between GRAPPA kernels and weight applications between GRAPPA and SPIRiT, depicted for a single coil. Source points are shown in blue (mid gray in print version), target points in orange (gray in print version), sampled points in dark gray, missing points in white, and estimated points in light gray. **(left)** In GRAPPA, weight sets are enforced only from sampled to nonsampled lines. **(right)** In SPIRiT, these weights are enforced over every patch of k-space. This allows for iterative improvement in the estimated values for the nonsampled points.

struction [35]. This property also makes CG-SENSE compatible with other iterative reconstruction techniques including compressed sensing, which will be introduced in Chapter 8.

As randomly/nonuniformly sampled Cartesian and non-Cartesian scans are generally not used clinically, CG-SENSE is mostly limited to research applications. However, it was the first major non-Cartesian parallel imaging technique, and it has been studied in cardiac [36] and fMRI [37] applications.

Iterative self-consistent parallel imaging reconstruction (SPIRiT)

In a mathematical sense, SPIRiT [4] is a generalization of the GRAPPA algorithm that considers GRAPPA as an iterative optimization problem. Much like GRAPPA, SPIRiT requires a calibration data set to determine reconstruction weights for a specific GRAPPA kernel that are then applied to recover uncollected k-space points in an undersampled data set.

The main idea behind SPIRiT is that the numerical relationships within a kernel should be satisfied over all kernels found in the fully sampled k-space data. This contrasts with the original GRAPPA algorithm, which only enforces the relationships captured in the GRAPPA weights to be satisfied between the originally sampled and non-sampled points. This principle is shown visually in Fig. 6.10. In SPIRiT, the GRAPPA weight operator, $\boldsymbol{G}$, is a convolution operation that is applied over the entire k-space data to enforce the relationships expected given the coil sensitivities. This operation can be expressed as

$$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x}. \tag{6.17}$$

Intuitively, this equation can be understood as stating that any point in k-space can be described as some combination of nearby k-space points using the GRAPPA operation. To reach this ideal solution and approximate the missing k-space values from an undersampled data set, this equation is posed as an optimization problem:

$$\operatorname{argmin}_{\boldsymbol{x}} \|(\boldsymbol{G}-\boldsymbol{I})\,\boldsymbol{x}\|_2^2, \tag{6.18}$$

where $\boldsymbol{I}$ is the identity matrix, and $\|\cdot\|_2$ indicates the L2-norm. This formulation ensures that the GRAPPA relationships are enforced over each local area of k-space.

While this approach could be applied to reconstruct accelerated data, it lacks a term to enforce data consistency and could converge to a solution that does not reflect the original data. For instance, $\boldsymbol{x}=0$ would satisfy the equation regardless of the GRAPPA operator deployed. A regularization term based on data consistency is thus included within the optimization structure

$$\operatorname{argmin}_{\boldsymbol{x}} \|(\boldsymbol{G}-\boldsymbol{I})\,\boldsymbol{x}\|_2^2 + \lambda\,\|\boldsymbol{B}\boldsymbol{x}-\boldsymbol{y}\|_{2,}^2 \tag{6.19}$$

where λ is a regularization parameter, $\boldsymbol{B}$ is a linear sampling operator that represents selection of the sampled k-space points, and $\boldsymbol{y}$ is the originally sampled data. In Cartesian sampling, $\boldsymbol{B}$ is a sampling mask matrix. SPIRiT can be extended to non-Cartesian datasets simply by using a gridding operation as the $\boldsymbol{B}$ matrix. Note that $\boldsymbol{x}$ is a Cartesian k-space data matrix (even for non-Cartesian datasets), and $\boldsymbol{B}$ does not typically include any Fourier transform operation. Additional regularization terms can be added to this objective function as desired. This problem can then be solved using any appropriate optimization algorithm, such as the conjugate gradient (CG) algorithm. More information on optimization algorithms can be found in Chapter 3.

The iterative nature of SPIRiT confers both benefits and drawbacks. Like CG-SENSE, regularization can easily be applied to SPIRiT reconstructions. For example, by incorporating L1-regularization to the wavelet transform of the image, the problem can be converted to a hybrid parallel imaging/compressed sensing reconstruction approach [38]. While iterative techniques tend to require more processing power compared to direct techniques such as SENSE and GRAPPA, the SPIRiT algorithm has been implemented with parallel CPU and GPU resources to enable rapid reconstruction on modern computers [38].

SPIRiT is not broadly used in clinical applications, but it has been applied in many types of research scans. For example, SPIRiT has been investigated for acceleration of pediatric scans, including cardiac, abdominal, and knee scans [39]. SPIRiT has also been investigated for potential applications in cardiac cine imaging [40].

Eigenvalue iterative self-consistent parallel imaging reconstruction (ESPIRiT)

The driving motivation behind the ESPIRiT algorithm is to balance the fundamental tradeoffs between k-space-based parallel imaging techniques such as GRAPPA or SPIRiT and coil sensitivity-based techniques such as SENSE. Autocalibrating techniques (like GRAPPA and SPIRiT) are typically more robust to noise and errors in sensitivity maps. However, sensitivity-based methods can have much better results in ideal circumstances and have other advantages, such as straightforward incorporation of regularization. ESPIRiT aims to combine the strengths of both approaches by using autocalibration data to robustly estimate coil maps for a coil-sensitivity-based approach, with improved reconstructions even in the presence of noise.

To achieve this goal, two new techniques are introduced within the SPIRiT framework as ESPIRiT [5]: (1) a novel method of robustly estimating coil sensitivity maps based on SPIRiT calibration data and (2) a variation of the SENSE algorithm that can utilize these eigenvector coil maps.

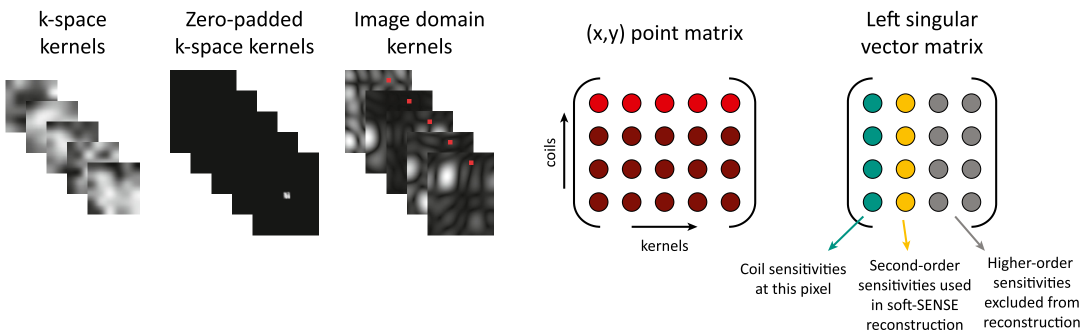

FIGURE 6.11

Diagram of coil estimation process using the ESPIRiT algorithm. First, a set of SPIRiT kernels are extracted from the k-space calibration data. These kernels are zero-padded to the desired coil map resolution and transformed into the image domain. Each image pixel has an associated matrix of size $N_c \times N_k$. The SVD of this matrix is taken, and the left singular vector matrix contains information about coil sensitivities. The first singular vector contains the primary coil sensitivity map values for that pixel, and the other singular vectors can contain higher-order coil sensitivity map values.

Like SPIRiT, ESPIRiT requires a fully sampled calibration dataset, from which a set of kernels is extracted in order to calculate robust coil sensitivity maps. ESPIRiT kernels are different from GRAPPA or SPIRiT kernels in that source and target points are not differentiated from one another. The entire k-space kernel is extracted from the calibration data and placed into a 4D calibration matrix of size $h_k \times w_k \times N_c \times N_k$, where h_k is the kernel height, w_k is the kernel width, N_c is the number of coils, and N_k is the number of kernels. Next, the calibration matrix is normalized, and the kernels are compressed using Principal Component Analysis (PCA) to reduce the number of kernels that must be stored in memory. The kernels are then zero-padded to the size of the image, and a 2D Fourier Transform is performed along the first two dimensions of the calibration matrix, resulting in a set of low-resolution kernel images. For each spatial coordinate (x, y), a corresponding matrix of size $N_c \times N_k$ with the information for each kernel and coil at that point is extracted. The singular value decomposition (SVD) of this matrix is calculated, resulting in a $N_c \times N_c$ left singular vector matrix and a set of N_c singular values. This process is shown in Fig. 6.11. This SVD can also be thought of as an eigendecomposition of a matrix associated with each pixel multiplied by its transpose, which is how the ESPIRiT algorithm derives its name. In this case, the eigenvector matrix is equivalent to the left singular vector matrix, and the eigenvalues are equivalent to the squared singular vectors. After PCA, the greatest variation in the signal along the coil dimension should be due to differences in coil sensitivities, and the most significant singular vectors should consequently contain the coil sensitivity values for that pixel. The most significant singular vectors have a corresponding singular value of 1 since all singular values are constrained to be less than or equal to 1. To create coil sensitivity maps, the first singular vectors are extracted for each pixel position and combined into a sensitivity map matrix of size $N_x \times N_y \times N_c$, where N_x and N_y are the image matrix dimensions. If a pixel's first eigenvalue is less than one, the zero vector is used instead of the first singular vector as it is assumed that the first singular vector is not related to coil sensitivities for this pixel. The resulting coil sensitivity maps can then be used in SENSE or CG-SENSE to reconstruct unaliased images.

In some cases, pixels can have multiple singular values equal to (or nearly equal to) one and multiple sets of associated coil sensitivity values. A second set of coil maps can be calculated in a process similar to that used for the first coil maps: The second singular vectors are extracted and combined into a map matrix, and pixels with a second eigenvalue less than one are set as the zero vector. This process can be repeated up to N_c times. There is value in including higher-order coil sensitivities during the reconstruction step as they can contain useful information missing in the first-order sensitivities. For example, if part of the imaged object lies partially outside the image field of view, the second-order map contains sensitivities for the aliased region. Neglecting to include these values can cause errors in the SENSE reconstruction. Two sets of coil maps are typically used, allowing for better reconstructions over traditional SENSE without requiring significantly additional processing time.

Like SPIRiT, ESPIRiT considers the reconstruction as an optimization problem. To perform an ESPIRiT reconstruction, an objective function incorporating multiple sets of coil maps is created:

$$\operatorname{argmin}_{\boldsymbol{x}} \sum_{i=1}^{N_c} \left\| \boldsymbol{B}\boldsymbol{F} \sum_{j=1}^{N_m} \boldsymbol{C}_{i,j}\boldsymbol{x} - \boldsymbol{s}_i \right\|_2^2, \tag{6.20}$$

where $\boldsymbol{s}_i$ is the sampled signal at the i^{th} coil, $\boldsymbol{B}$ is the sampling operator, $\boldsymbol{F}$ is the Fourier transform operation, N_m is the number of coil sensitivity map sets used for reconstruction, $\boldsymbol{C}_{i,j}$ is the i^{th} coil from the j^{th} coil set, and $\boldsymbol{x}$ is the image. In the case in which there is only one set of coil sensitivity maps ($N_m = 1$), this becomes equivalent to the CG-SENSE algorithm or the traditional SENSE algorithm in the Cartesian case.

Like SPIRiT, ESPIRiT is not yet used routinely for clinical scans. The ESPIRiT coil map-estimation process [41] has been adopted for some research applications, and recent investigations have also used the ESPIRiT algorithm for cine imaging [42].

6.4 3D volumetric parallel imaging

While the techniques discussed so far have been focused on the reconstruction of 2D images, parallel imaging can also be used to effectively accelerate the acquisition of 3D volumetric scans. For such scans, the typical convention is to accelerate in both the phase-encoding and partition-encoding (second phase encoding) directions. By accelerating the acquisition along two dimensions, higher acceleration factors can be deployed due to the orthogonality of the coil-encoding capabilities in various directions. As with 2D applications, undersampling in the frequency encoding (readout) direction yields minimal time savings.

In the volumetric extension of SENSE (called 2D SENSE since undersampling occurs in two directions) [6], the SENSE equation (Eq. (6.6)) is still applicable. For example, a dataset undersampled by R = 2 in both the phase-encoding and partition-encoding directions has a total acceleration factor of R = 4. As a result of this sampling scheme, four pixels are aliased into one single pixel in the aliased image; two replicas are seen in the phase-encoding direction and two in the partition-encoding direction. The effects of this undersampling scheme are shown in Fig. 6.12. As before, the SENSE equations

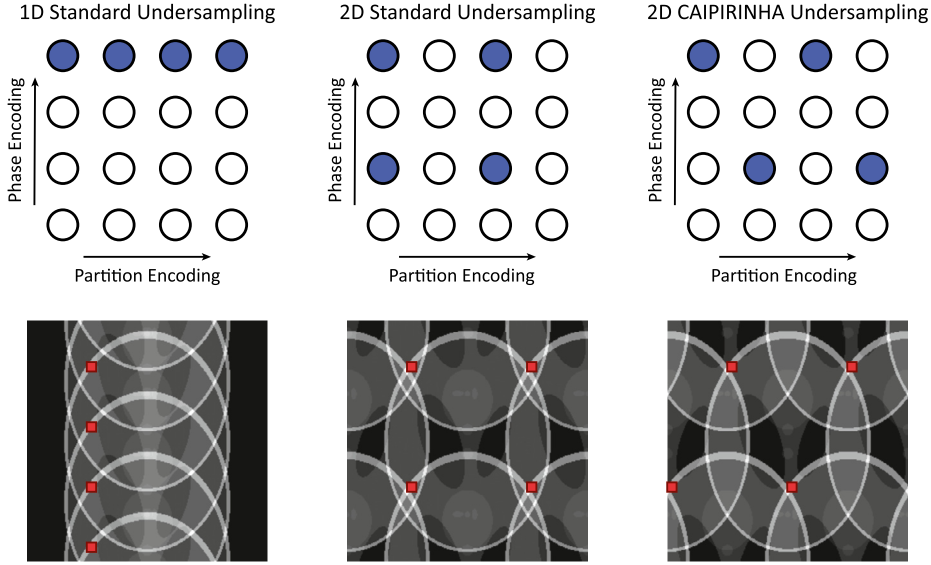

FIGURE 6.12

An example of using different sampling schemes with a total acceleration factor of R = 4. **(left)** Sampling and aliasing patterns for 1D undersampling along the phase-encoding direction. In this image, many areas near the center of the image have three or four overlapping signal-bearing pixels. **(center)** Using an undersampling pattern of R = 2 in both the phase-encoding and partition-encoding directions lowers the amount of overlapping signal in many parts of the image. However, there are still areas in this image where four signal-bearing pixels overlap. **(right)** In the CAIPIRINHA scheme, the maximum number of overlapping signal-bearing pixels is only three, and the total overlapped area is reduced in comparison to the standard 2D undersampling scheme. The reduction in overlapping information leads to a lower g-factor in the reconstructed images and thus less noise and fewer residual aliasing artifacts.

are structured so that the aliased points are the sum of the unaliased pixels multiplied by the coil sensitivity at that pixel location. The coil sensitivity coefficient matrix can be inverted, and the unaliased pixel values can be found.

In 2D GRAPPA [7] (also named for the number of undersampled dimensions, not the number of spatial dimensions), kernels can be defined across three spatial dimensions. An example of a 3D GRAPPA kernel is shown in Fig. 6.13. The remainder of the 2D GRAPPA reconstruction is largely equivalent to standard GRAPPA. The source and target matrices will have more columns since 3D kernels typically have more points than 2D kernels. Once calculated, the GRAPPA weights can be applied over all kernels in the 3D volume to reconstruct the missing data.

Highly accelerated 3D imaging is often facilitated by the application of Controlled Aliasing in Parallel Imaging Results in Higher Acceleration (CAIPIRINHA) [8], a sampling technique in which sampled lines are offset in the phase-encoding and partition-encoding directions, as shown in Fig. 6.12. When using such a sampling scheme, aliasing artifacts are shifted in the phase/partition directions, so there is less overlap in the aliased images. The reduction in signal aliasing leads to a lower g-factor in

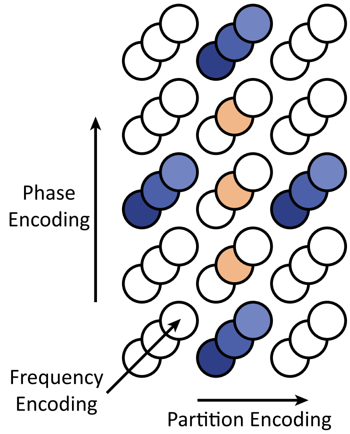

FIGURE 6.13

Example of a GRAPPA kernel defined in three dimensions, with an acceleration factor of R = 4 and a CAIPIRINHA sampling scheme. The source points are shown in blue (dark gray in print version) and target points in orange (light gray in print version). Note that unlike Fig. 6.7, only a single coil is shown here for ease of presentation, but the GRAPPA kernel would actually be made up of source points across multiple coils.

parts of the image and improves image quality. This sampling scheme can be used in conjunction with any parallel imaging algorithm designed to work on 3D datasets.

3D parallel imaging techniques are used clinically alongside traditional GRAPPA and SENSE. Some common applications for 3D imaging techniques include MR angiography [27] and musculoskeletal MRI [43], among many others.

6.5 Dynamic parallel imaging

The previous sections describe parallel imaging approaches that can be applied to a single static image. Dynamic applications, such as cardiac cine imaging, can also greatly benefit from parallel imaging to improve temporal resolution without significant compromises in image quality. While previously discussed approaches can be applied frame-wise in dynamic imaging, it is often beneficial to incorporate temporal information in the reconstruction. The following sections describe a few of these notable dynamic parallel imaging approaches: TSENSE and k-t SENSE, TGRAPPA, PEAK-GRAPPA, and through-time GRAPPA.

TSENSE and k-t SENSE

Similar to the CAIPIRINHA sampling scheme, in which data are collected in an offset pattern, many temporal parallel imaging algorithms require data sampled in an offset pattern in the temporal dimension. For example, with an acceleration factor of R = 2, odd phase-encoding lines would be sampled for the first temporal frame, even phase encoding lines in the second frame, and so on. This sampling scheme introduces aliasing along one spatial dimension and the temporal frequency dimension, which

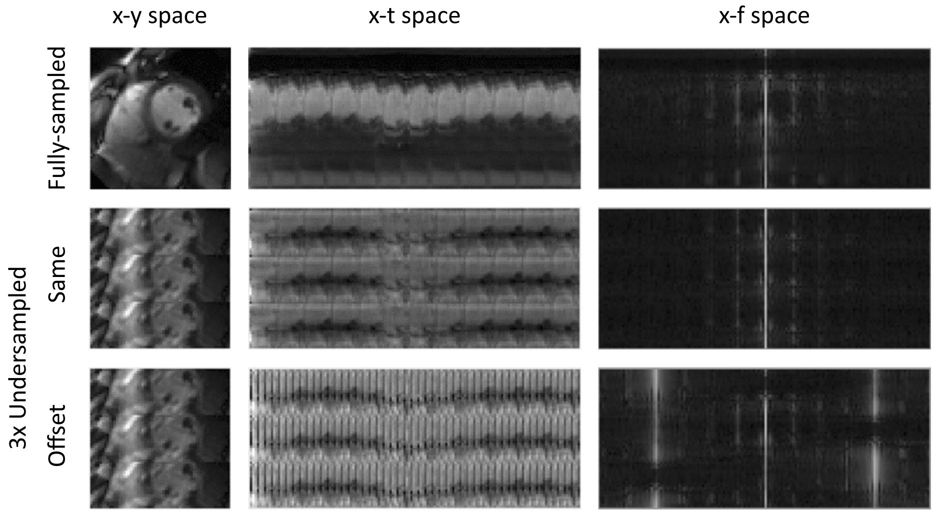

FIGURE 6.14

Example of cardiac MRI data, shown in different domains. **(top)** Representations of fully sampled data, without any aliasing artifacts, shown for reference. **(center)** R = 3 undersampled data, where the same lines of k-space data are collected for each frame. **(bottom)** R = 3 undersampled data, where the sampled lines are alternated in each frame as described in the text. **(left)** Image domain representations of cardiac data. **(center)** x-t space representation, showing action along a single line through time. Note the "flickering" in the offset undersampled x-t space that arises from the interleaved sampling pattern. **(right)** x-f space representations generated by calculating the temporal Fourier transform of the x-t representations. In the R = 3 undersampled x-f space, multiple replicates of the x-f space signal can be seen, where each replica is slightly shifted, again due to the sampling pattern. When the same k-space line is sampled, these replicates are only shifted along the x dimension and are aliased over the original spectrum. When sampled in the offset pattern, these replicates are shifted in both spatial (x) and frequency (f) dimensions, resulting in significantly less overlap between replicates in the x-f domain.

reduces the amount of signal overlap occurring in the spatial position/temporal frequency domain, also known as x-f space. An example of information in the x-f space is shown in Fig. 6.14. Since the temporal frequency domain is relatively sparse, higher acceleration factors can be deployed before significant aliasing occurs.

TSENSE [12] reconstructs data in a two-step process, combining the SENSE and UNFOLD [44] algorithms. The SENSE algorithm is first applied to each undersampled frame in the image domain, reducing spatial aliasing in the data set. A temporal low-pass filter is then applied to the data, reducing temporal aliasing effects. Between these two methods for reducing aliasing artifacts, higher acceleration factors can be achieved.

k-t SENSE [13] works in a similar manner and is an extension of the k-t BLAST algorithm described in Chapter 5. The k-t BLAST algorithm uses initial estimates of signal strength to determine the most effective filter to remove aliasing in the x-f space. At each point in the aliased image, an estimate is calculated for the fraction of the signal attributed to each signal source point. The aliased pixel signal is then redistributed to reconstruct pixels based on these estimated signal fractions.

k-t SENSE improves upon this algorithm by incorporating coil sensitivity profiles into the estimated signal strength matrix. Much like traditional SENSE, k-t SENSE is performed pixel-wise on the aliased image. This can be achieved by inserting the coil sensitivity matrix into the k-t BLAST equations

$$\hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}_0 + \boldsymbol{M}^2 \boldsymbol{C}_p^{\mathrm{H}} \left(\boldsymbol{C}_p \boldsymbol{M}^2 \boldsymbol{C}_p^{\mathrm{H}} \right)^{-1} \left(\boldsymbol{x}_{alias} - \boldsymbol{C}_p \hat{\boldsymbol{x}}_0 \right) \tag{6.21}$$

In this equation, $\boldsymbol{x}_{alias}$ is the aliased image space signal vector of length N_c (# of coils), $\hat{\boldsymbol{x}}$ is the resulting estimated signal vector of length R (the acceleration factor), $\hat{\boldsymbol{x}}_0$ is an initial estimate vector of length R, $\boldsymbol{M}^2$ represents the $R \times R$ covariance matrix of signal deviation from $\hat{\boldsymbol{x}}_0$, and $\boldsymbol{C}_p$ is the $N_c \times R$ encoding matrix of coil sensitivity values used in the traditional SENSE reconstruction. This equation can be solved for each point in the aliased image to reconstruct the dynamic data.

In addition to k-t SENSE, the CG-SENSE algorithm can also be adapted for use in non-Cartesian dynamic imaging applications. For dynamic CG-SENSE, regularization terms can be added to the CG-SENSE cost function to enforce certain image properties over time. For example, Otazo, et al. [45] used spectral sparsity (i.e., L1-norm of the temporal Fourier transform) as a regularizer for cardiac perfusion data. More information on this type of technique can be found in Chapter 8.

One of the greatest benefits of using temporal methods for dynamic image reconstructions is that they exploit redundancies in the time domain. In this case, the temporal frequency domain is relatively sparse and most information is contained near the f = 0 point. This sparsity allows for a high degree of undersampling before temporal signals alias in the frequency domain, resulting in improved reconstructions at higher acceleration factors than SENSE or k-t BLAST alone. k-t SENSE is commonly used in place of SENSE for applications in dynamic imaging, such as cardiac cine [46] or myocardial perfusion [47].

TGRAPPA, k-t GRAPPA, & PEAK-GRAPPA

A few different variants of the GRAPPA algorithm have been created specifically for dynamic imaging. Three of these techniques are TGRAPPA, k-t GRAPPA, and Parallel MRI with Extended and Averaged Kernels GRAPPA (PEAK-GRAPPA).

TGRAPPA [14] is the most straightforward of the three as it works with the same kernel design as traditional GRAPPA. Much like k-t BLAST/SENSE, the phase-encoding lines that are collected are interleaved and alternated for each frame. After collecting a series of R frames, the data can then be combined to create a fully sampled k-space, which is used as the ACS data set, as shown in Fig. 6.15a. GRAPPA weights are determined using this temporally combined data and then applied to the undersampled frames to reconstruct the missing data points for each frame. This approach enables imaging with a higher frame rate without the need to collect dedicated ACS data.

k-t GRAPPA [15] uses a slightly different approach for reconstructing temporal data. k-t GRAPPA uses a similar offset phase-encoding sampling pattern as TGRAPPA, but uses a different type of GRAPPA kernel. Rather than just using spatial k-space neighbors, k-t GRAPPA also includes the two nearest temporal neighbors in the GRAPPA kernel as well. A diagram of this sampling scheme is shown in Fig. 6.15b. k-t GRAPPA can be calibrated using either a dedicated ACS dataset, in the form of several central k-space lines collected along with each frame, or using a TGRAPPA calibration scheme where the ACS is generated from a series of combined frames from an offset sampling pattern.

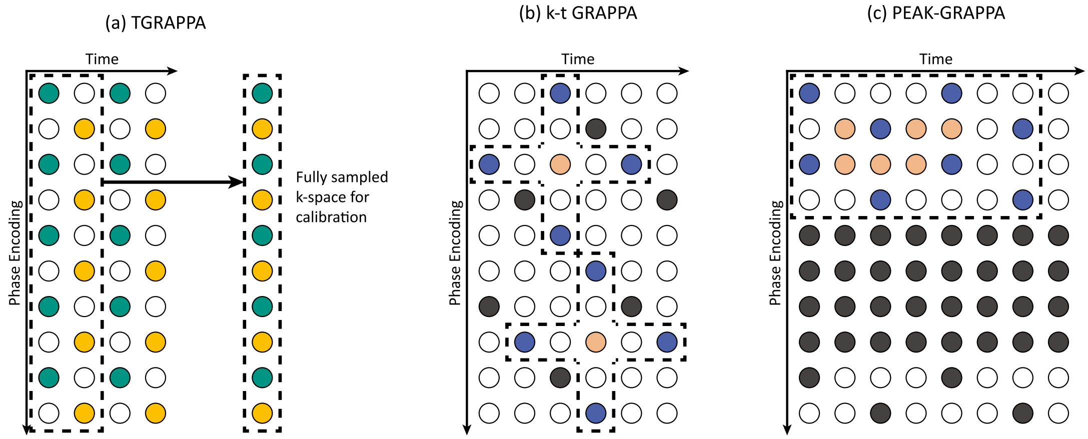

FIGURE 6.15

(a) TGRAPPA sampling. Different interleaves of undersampled data are collected at each time frame. If data are combined over R of these time frames, the result is a fully sampled dataset that can be used for calibration. **(b)** k-t GRAPPA sampling and kernel structure. In k-t GRAPPA, kernels include source points in k-space from the same time frame, as well as adjacent time frames. Kernels contain a single target point (orange (light gray in print version)), and the closest neighbors both in time and in k-space are used as source points (blue (mid gray in print version)) **(c)** PEAK-GRAPPA sampling and kernel definition. A sample kernel is shown near the top of the figure, with source (blue (mid gray in print version)) and target (orange (light gray in print version)) points. Sampled points outside this kernel are shown in dark gray. Unlike TGRAPPA, central calibration lines are still required since temporal information is needed to properly calibrate the temporal kernels.

PEAK-GRAPPA [16] is a more generalized form of k-t GRAPPA. Rather than only using a single target point and its four nearest spatiotemporal neighbors as source points, PEAK-GRAPPA uses a larger kernel. A diagram of a PEAK-GRAPPA kernel and sampling scheme is shown in Fig. 6.15c. PEAK-GRAPPA also includes the collection of a dedicated ACS dataset to enable accurate characterization of kernel time dynamics.

Temporal GRAPPA approaches can be used in place of GRAPPA for dynamic applications since temporal redundancies can allow for higher acceleration factors. Like k-t SENSE, this includes applications such as cardiac cine [48] [49] or perfusion imaging [50].

Through-time non-Cartesian GRAPPA

Through-time non-Cartesian GRAPPA [51] is a method for applying dynamic parallel imaging to non-Cartesian data sets. Through-time non-Cartesian GRAPPA uses a different approach compared to other dynamic parallel imaging techniques: Rather than using temporal data to reduce x-f space aliasing, through-time non-Cartesian GRAPPA uses temporal information to increase the amount of data available for the calibration step.

Several variations of the GRAPPA algorithm exist for non-Cartesian sampling schemes [10] [11]. When working with non-Cartesian data, there are some additional considerations that must be made due to the sampling geometry [52]. In Cartesian GRAPPA, the shape of the GRAPPA kernel is constant

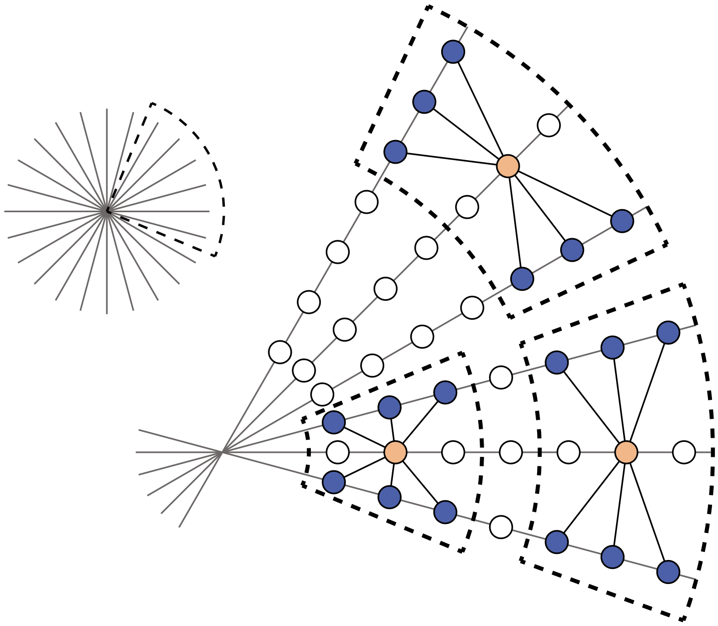

FIGURE 6.16

GRAPPA kernel definition in radial k-space data. In radial GRAPPA, kernel geometries are not consistent across all of k-space. Distances between the points in kernels near the center of k-space are much smaller than distances between points in kernels near the edge of k-space, and so there is no longer a single weight set that can be applied to both kernels to accurately reconstruct the missing points. Rotations in k-space also distort the positioning between neighboring points, and so separate weight sets must be used for each readout line.

throughout k-space, which allows weights to be generated from a large ACS dataset for the single kernel geometry and applied uniformly across all of k-space. In non-Cartesian GRAPPA, the shape of the GRAPPA kernel is not constant and shifts depending on the positions of the sampled k-space points. Thus, a single kernel shape cannot be used to describe the relationship between all subsets of collected and missing k-space points, as shown in Fig. 6.16. Despite this challenge, there are a few ways to deploy GRAPPA for non-Cartesian data. In general, formulations of non-Cartesian GRAPPA assume that kernel geometries are constant within small segments of non-Cartesian k-space. Weights can then be calculated for each segment of k-space from a fully-sampled non-Cartesian dataset and then applied locally to the undersampled data.

While GRAPPA can be used to generate an image from undersampled non-Cartesian data, the need for a fully sampled dataset for calibration reduces the efficiency of the acquisition. However, in the case of dynamic imaging, where the calibration time is less important than the temporal resolution of the accelerated images, non-Cartesian GRAPPA can provide high frame rates with low levels of residual aliasing artifacts. In through-time non-Cartesian GRAPPA [10] [11], several frames of fully-sampled k-space are collected prior to a dynamic MRI scan. GRAPPA weights are then calculated over very small segments of k-space, primarily relying on multiple kernel repetitions through time to generate stable GRAPPA weights. Once GRAPPA weights are generated for these local segments, they can be applied to a dynamic series of highly-accelerated non-Cartesian frames to achieve high temporal resolution.

Through-time non-Cartesian GRAPPA has been applied to accelerate dynamic imaging with high frame rate requirements, such as cardiac [51] and perfusion [53] imaging, where higher temporal resolution can capture rapid motion or contrast changes. In addition, through-time non-Cartesian GRAPPA reconstructions can be performed rapidly for real-time imaging applications [54].

6.6 Artifacts in parallel imaging

While parallel imaging is used in nearly all clinical scans to accelerate data collection, there are a number of artifacts that can arise when the parallel imaging algorithm fails to completely resolve the aliasing due to undersampling. Noise enhancement is one common artifact seen in parallel imaging. While some decrease in SNR is expected with parallel imaging, this artifact refers to a dramatic decrease in SNR in a specific region of an image. This artifact is often seen in image areas where many pixels alias together, and the coil sensitivity profiles contain less distinct values (resulting in ill-conditioning of the matrix and thus noise amplification during inversion) [29]. This situation typically occurs near the center of images, where coil sensitivity maps can have relatively similar values. This can cause the SENSE coil sensitivity encoding matrix or GRAPPA ACS source point matrix to be nearly singular, thereby increasing the g-factor of the region. Some examples of this artifact are shown in Figs. 6.17 and 6.19. Noise enhancement artifacts can typically be resolved by reducing the acceleration factor [28].

Residual aliasing occurs as a result of poor estimation of SENSE coil sensitivity maps or GRAPPA weights [29]. In this case, errors in the sensitivity information can cause some aliasing artifacts to remain in the image even after reconstruction. This situation can occur in cases where the patient moves between the coil sensitivity estimation or ACS collection and the actual scan. This motion causes the coils to be loaded differently, changing their sensitivity profiles [1]. In GRAPPA, this can also occur if the ACS is chosen to be too small, which can cause inaccuracies in the GRAPPA weights [28]. Examples of residual aliasing artifacts are shown in Figs. 6.18 and 6.19. Residual aliasing can also occur alongside noise enhancement because singularity in the coil sensitivity matrix or source point matrix introduces errors in the reconstruction, which can manifest as residual aliasing. Residual aliasing artifacts can typically be resolved by recollecting or increasing the amount of calibration data [28].

6.7 Summary

Parallel Imaging is a class of reconstruction algorithms in which multiple receiver coils are used to resolve aliasing artifacts intentionally generated by undersampling k-space. These techniques can be used to greatly reduce the amount of time required to collect many different types of MRI scans or improve spatial resolution with no additional required imaging time. The most commonly used techniques include SENSE, GRAPPA, SPIRiT, and ESPIRiT, each of which have been used to accelerate MRI scans in the clinic. Variations of these techniques have been developed to efficiently reconstruct accelerated nonuniform Cartesian, non-Cartesian, volumetric, and dynamic data sets. Moreover, in recent years, parallel imaging has been successfully combined with methods that exploit other sources of

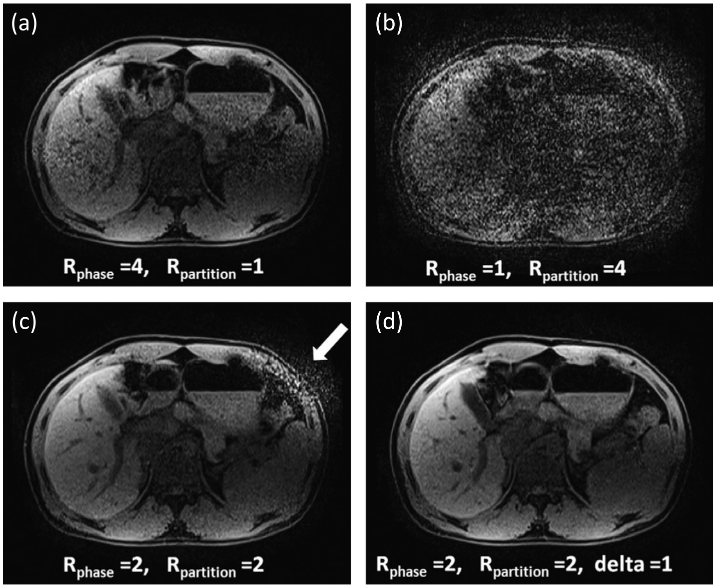

FIGURE 6.17

Example of noise enhancement in a 3D abdominal scan reconstructed with 2D GRAPPA, originally published in [28]. **(a)** Image showing noise enhancement artifacts due to excessive oversampling in the phase-encoding direction. **(b)** Image showing noise enhancement artifacts due to oversampling in the partition-encoding direction. Partition-encoding artifacts are worse than the phase-encoding artifacts because the coil sensitivities were similar along the partition-encoding direction compared to the phase-encoding direction. **(c)** Image showing a reconstruction that used undersampling along both phase-encoding and partition-encoding directions. This sampling scheme resolves most of the noise-enhancement artifacts, but there is still a small region where the coils do not support reconstruction for this undersampling scheme. **(d)** Image reconstructed using a CAIPIRINHA sampling scheme that effectively removes the artifacts and reduces noise enhancement.

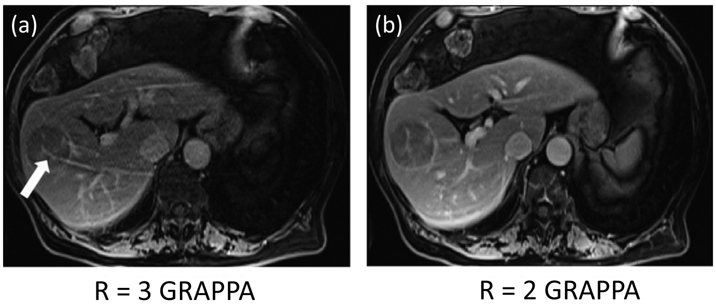

FIGURE 6.18

Example of residual aliasing artifact in an abdominal scan reconstructed with GRAPPA, originally published in [28]. **(a)** Image with an acceleration factor of R = 3. In this case, residual aliasing can be seen near the center of the image. **(b)** Image with an acceleration factor of R = 2. In this case, the residual aliasing artifact is cleared by reducing the acceleration factor to a level that the coil geometry can support.

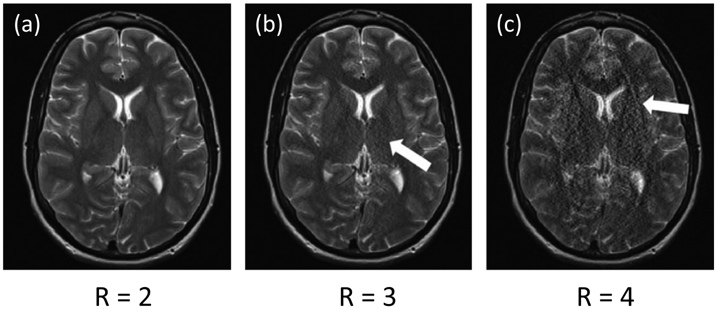

FIGURE 6.19

Example of residual aliasing co-occurring with noise enhancement in a brain scan reconstructed with GRAPPA, originally published in [28]. As the acceleration factor is increased, noise enhancement and residual aliasing artifacts appear since the coil geometry is no longer able to support an accurate parallel imaging reconstruction.

prior information and regularization (for example, low-rank reconstructions and compressed sensing); these combined techniques will be described in the following chapters.

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